HCNet: Hierarchical Feature Aggregation and Cross-Modal Feature Alignment for Remote Sensing Image Captioning

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***Abstract*— Remote sensing image captioning (RSIC) aims to describe the crucial objects from remote sensing images in the form of natural language. The inefficient utilization of object texture and semantic features in images, along with the ineffective cross-modal alignment between image and text features, are the primary factors that impact the model to generate high-quality captions. To alleviate this trouble, this article presents a network for RSIC, namely HCNet, including hierarchical feature aggregation and cross-modal feature alignment. Specifically, a hierarchical feature aggregation module (HFAM) is proposed to obtain a comprehensive representation of vision features, which is beneficial for producing accurate descriptions. Considering the disparities between different modal features, we design a cross-modal feature interaction module (CFIM) in the decoder to facilitate feature alignment. It can fully utilize cross-modal features to localize critical objects. Besides, a cross-modal feature align loss is introduced to realize the alignment between image and text features. Extensive experiments show our HCNet can achieve satisfactory performance. In particular, we demonstrate significant performance improvements of +14.15% CIDEr score on NWPU datasets compared to existing approaches. The source code is publicly available at https://github.com/CVer-Yang/HCNet.**

***Index Terms*— Attention mechanism, feature aggregation, feature alignment, image caption, remote sensing.**

I. Introduction

REMOTE sensing image captioning (RSIC) is a comprehensive task that combines natural language processing and computer vision, which has attracted extensive interest due to its significant application, such as image retrieval [2], scene understanding [3], change detection [4], and other fields. Compared with tasks such as image super-resolution [5], [6], object detection [7], and semantic segmentation [8], image captioning not only recognizes objects in the image but also captures the relationships of objects.

RSIC faces several special challenges.

1) *Scale variations:* objects with the same category in remote sensing images may display considerable scale variations, for example, an airport may contain airplanes of different sizes. To generate accurate and meaningful captions, RSIC models need to integrate features across different scales and capture the relationship between them.
2) *Cross-modal disparities:* given a remote sensing image, the model generates corresponding textual descriptions. There exist substantial modality differences between the input and output, requiring precise alignment of image and text features during the text generation and achieving feature fusion effectively. The above challenges have a significant impact on producing high-quality captions, which makes the RSIC more difficult than other remote sensing tasks.

The RSIC models typically can be split into two stages: vision feature extraction and text generation. In the vision feature extraction stage, these methods commonly employ convolutional neural networks (CNNs) [9] or vision Transformer [10] to extract features from images and encode them into high-dimensional feature vectors. In the text generation stage, recurrent neural networks or Transformer [11] are utilized to transfer the high-dimensional feature vectors into corresponding text. Currently, most methods design the multiscale features integration, several attention mechanism combinations [15], [16], and auxiliary tasks [17] to construct the models for text generation. Although these approaches have achieved promising results, the following problems still exist: 1) in the vision feature extraction stage, these feature fuse methods [12], [13], [14] cannot effectively utilize the texture and semantic information in different scale features, which results in inaccurate category descriptions in the generated captions and 2) during the text generation stage, the multimodal features are directly fed into the long short-term memory (LSTM) network without effective alignment, which will cause semantic confusion and limit the ability of model to accurately locate the objects. Therefore, *how to effectively* ***obtain comprehensive vision representation*** *and* ***align cross-modal features*** *is significant for the RSIC task.*

To alleviate these problems, this article proposes a novel method for RSIC called HCNet, which utilizes hierarchical feature aggregation and cross-modal feature alignment. The model aims to acquire comprehensive visual features and produce precise hidden states. Firstly, we employ ResNet50 [18] as the encoder to extract features at multiple resolutions.

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This allows the model to implicitly capture rich and detailed information from the input images. To further strengthen the feature representation capability, a hierarchical feature aggregation module (HFAM) is proposed to effectively mine both texture and semantic features. During the decoder stage, we utilize a dual-LSTM architecture to generate coherent and contextually relevant captions. Additionally, a cross-modal feature interaction module (CFIM) is designed to facilitate the feature alignment from different modalities. The module enables effective information exchange of remote sensing images and text representations, leading to improved caption generation. Furthermore, a cross-modal feature alignment loss is introduced to enhance the quality of the produced captions. This loss guides HCNet to mitigate the disparity between image and text features. The main contributions of this work are summarized as follows.

1) We propose an HCNet for the RSIC task, which incorporates two key modules: HFAM and CFIM. We verify that the proposed model with the two main modules exhibits superior performance, which reveals the effectiveness of the two modules for model learning and optimization. Compared with the previous RSIC methods, the proposed HCNet can produce accurate and complete captions.
2) The HFAM with texture completion submodule and semantic distribution submodule is proposed, which can provide rich structure features from shallow features and precious semantic features from deep features to reduce caption confusion.
3) The CFIM is designed to enhance the alignment between the image and different modal features, thereby enabling the model to generate accurate hidden states. Besides, we introduce a cross-modal feature align loss to minimize the differences between image and text features.

The rest of this article is organized as follows. Section II provides related work in natural image captioning (NIC) and RSIC. In Section III, we describe the details of our proposed HCNet. The experimental results are analyzed in Section IV. Finally, Section V offers the conclusion.

II. Related Work

In this section, we review the existing image captioning methods. They are roughly divided into NIC and RSIC.

A. Natural Image Captioning

Currently, the mainstream NIC methods for caption generation are usually built by using the encoder–decoder framework. Here, the encoder is employed to extract features and the decoder generates corresponding text descriptions. For instance, Vinyals et al. [19] utilize CNN to convert images into visual features, and these features are then input into the LSTM network for text generation. Anderson et al. [20] introduce a dual-LSTM decoder, where Att-LSTM focuses on important image content and L-LSTM is responsible for word generation. It results in a substantial improvement in caption quality. Later, some researchers combine attention mechanisms to enhance the caption quality. Xu et al. [21] integrate a positional attention mechanism into the decoder stage to selectively focus on information from different positions during word generation. Additionally, Chen et al. [22] introduce an attention mechanism into the encoder and leverage key image features to generate words. Similar methods have [23] and [24]. Inspired by the benefits of Transformer, a collaborative network is proposed in [25] to realize the fusion of region and grid features for the captioning task. Wang et al. [26] utilize Swin Transformer [10] and Transformer respectively to extract image features and generate words. Kuo and Kira [27] employ the shared encoder to obtain object features, grid features, and text features of images, while the hierarchical decoder is designed to weigh these input features. This method achieves an obvious enhancement in CIDEr. Luo et al. [28] develop a semantic diffusion network, and adopt a pretrained retrieval model to obtain descriptions related to visual content. This approach demonstrates the potential of the diffusion model in image captioning. Ramos et al. [29] propose a caption generation framework that combines retrieval techniques, the model is light to train and has perfect domain adaptation capabilities. Ma et al. [30] combine the diffusion model with large language model (LLM) techniques to create a synthetic image–text dataset that improves the ability of the model to describe images from different perspectives. Although these methods have made progress in the description of natural images, their performance can be significantly degraded for complex remote sensing images when applied directly. In this case, we can refer to these methods to design corresponding algorithms for RSIC.

B. Remote Sensing Image Captioning

There are some differences between NIC and RSIC, including data source, content characteristics, and linguistic representation. In the past decade, some improved RSIC methods have been proposed [31]. For instance, Huang et al. [12] combine the denoising strategy with feature fusion to improve the caption quality. Ma et al. [14] propose two feature fusion modules and introduce object detection to obtain rich visual feature representation. Similar methods have [13] and [42]. To integrate the cross-modal features accurately, Zhang et al. [15] propose the linguistic state-guided attention in the decoder to remove irrelevant information in the fused feature maps. Similarly, Ye et al. [17] design a multilabel classification task to capture prior knowledge, and a semantic gate module is developed to steer the generation of hidden states. To enhance the interpretability of the model, a word-sentence framework is developed in [32], which first extracts words from images and then generates sorted description sentences by Transformer. Hoxha and Melgani [33] propose an SVM-based decoder that converts visual vectors into sentences and can generate high-quality textual descriptions in small datasets. Additionally, researchers also propose some methods based on Transformer, such as Chen et al. [34] construct the Swin Transformer network to explore features of images and use the Transformer network as a decoder to generate description sentences. To alleviate the overfitting problem, a truncation cross-entropy loss is proposed in [35] and achieves superior performance on small datasets. Lu et al. [36] take the

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audio input as active attention to generate accurate captions. Zhao and Xiong [37] provide regional-level annotations for remote sensing caption datasets and propose a method that fully utilizes regional features with grid features. Recently, Hu et al. [38] employ LLM techniques to explore the task of RSIC. Although existing RSIC methods have yielded satisfactory performances, they are still unable to obtain accurate captions. On the one hand, these methods fail to efficiently leverage the spatial information and semantic features in the fused feature maps. Therefore, the models cannot produce a comprehensive visual feature representation. On the other hand, these methods [15], [17], [34] do not fully aggregate cross-modal features in the decoder, which results in the generation of inaccurate hidden states. These factors make it difficult to generate accurate and complete captions.

III. Proposed Method

A. Overview

In remote sensing images, objects usually display significant scale variations. Besides, there exist substantial cross-modal disparities between remote sensing images the input and output for RSIC tasks. To overcome these challenges, we concentrate on improving this task in crucial aspects: hierarchical feature aggregation and cross-modal feature alignment. Fig. 1 shows the overall framework of HCNet. First, the pretrained ResNet50 on the ImageNet dataset is employed as the encoder to extract features. It yields four feature maps with different resolutions, where they are denoted as E1, E2, E3, and E4, respectively. Subsequently, the feature maps E2, E3, and E4 are input into the HFAM, which obtains a comprehensive visual feature representation. Unlike how previous approaches often concatenate the generated visual features, word features, and LSTM hidden states directly and feed the fusion features into the LSTM network, we recognize the significant disparities in remote sensing images with different modal features and design a CFMI that aligns visual features with different input features. Meanwhile, the cross-modal feature align loss is introduced to minimize the distance between remote sensing images the visual features produced by HFAM, and the sentence features encoded by LSTM. Finally, the aligned visual features, word features, and LSTM hidden states are embedded in the dual-LSTM network with Att-LSTM and L-LSTM to produce image captions.

B. Hierarchical Feature Aggregation Module

In contrast to natural images, remote sensing images typically capture a broad geographical area from a top-down remote perspective, where ground objects vary greatly in size and shape. It means that objects in images have small sizes, which makes it difficult to determine the category of objects by their appearance alone. It tends to obtain inaccurate feature representation. Interestingly, the surrounding environment provides a valuable complement to the shape, orientation, and other features of objects. A natural manner is to combine local texture features and global semantic information to build the model. To this end, an HFAM is designed to generate accurate vision features. The architecture of the HFAM is illustrated in Fig. 2. Specifically, we utilize several large kernels [39] with 15×15, 11×11, and 7×7 to explore the long-distance context from E2, E3, and E4. Subsequently, the feature maps are resized to the same size as E3 with 512 channels. To effectively fuse the multiscale features, the *texture completion submodule* and *semantic distribution submodule* are developed. The process is formulated as

$$E_i' = \mathrm{Conv}(E_i), \quad i \in \{2, 3, 4\} \tag{1}$$

where $\mathrm{Conv}(\cdot)$ is convolution operation.

1) Texture Completion Submodule: High-resolution feature map E_2' contains abundant detail features, which is conducive to helping the model identify salient objects. To obtain the position weights of the hidden feature map, an average pooling operation on the channel dimension is employed, which is followed by a Sigmoid operation. The position weight is then utilized in a matrix multiplication to fuse E_2' and E_3'. By doing so, the strategy achieves the information integration from the two feature maps according to position attention. It strengthens the representation of texture features in the fused feature maps. The process can be defined as

$$F_{\mathrm{detail}} = \mathrm{Sigmoid}\big(\mathrm{Avg}\big(E_2'\big)\big) \otimes E_3' \tag{2}$$

where $\mathrm{Avg}(\cdot)$ represents the average pooling and $\otimes$ denotes matrix multiplication.

2) Semantic Distribution Submodule: Low-resolution feature map E_4' can capture robust semantic information. It is conducive to learning the relationships of remote sensing image objects in the decoder. For that reason, we adopt global average pooling to condense the position feature. As a result, it obtains the channel weights of the hidden feature maps after two full connection (FC) layers and a Sigmoid operation. Finally, E_3' and E_4' are merged by matrix multiplication operation to yield semantic distribution. The process of semantic distribution submodule is represented as

$$F_{\mathrm{sem}} = \mathrm{Sigmoid}\big(\mathrm{FC}\big(R\big(\mathrm{FC}\big(\mathrm{Avg}\big(E_4'\big)\big)\big)\big)\big) \otimes E_3' \tag{3}$$

where $\mathrm{FC}(\cdot)$ is FC operations, and $R(\cdot)$ is a rectified linear unit (ReLU). Then we contact the texture feature map F_{detail} and the semantic feature map F_{sem}, and a convolution operation is utilized to conduct feature fusion. This process produces a visual feature map F_{vision} with the comprehensive feature representation. Finally, the vision feature map is reshaped to $\overline{V} \in \mathbb{R}^{196\times 1024}$ by using an Avgpooling and Flatten operation, that is,

$$F_{\mathrm{vision}} = \mathrm{Conv}([F_{\mathrm{detail}}, F_{\mathrm{sem}}]) \tag{4}$$

$$\overline{V} = \mathrm{Flatten}(\mathrm{Avg}(F_{\mathrm{vision}})) \tag{5}$$

where $[,]$ is the concatenation operation.

C. Cross-Modal Feature Interaction Module

During the caption generation stage, the vision features $\overline{V}$ obtained by HFAM, the previous hidden state h_{t-1} from the LSTM network, and the word feature F_{word} at the time t are input into the LSTM simultaneously to generate hidden states h_t. Here, h_{t-1} contains the text semantic information generated by the model before time t, while F_{word} has word information

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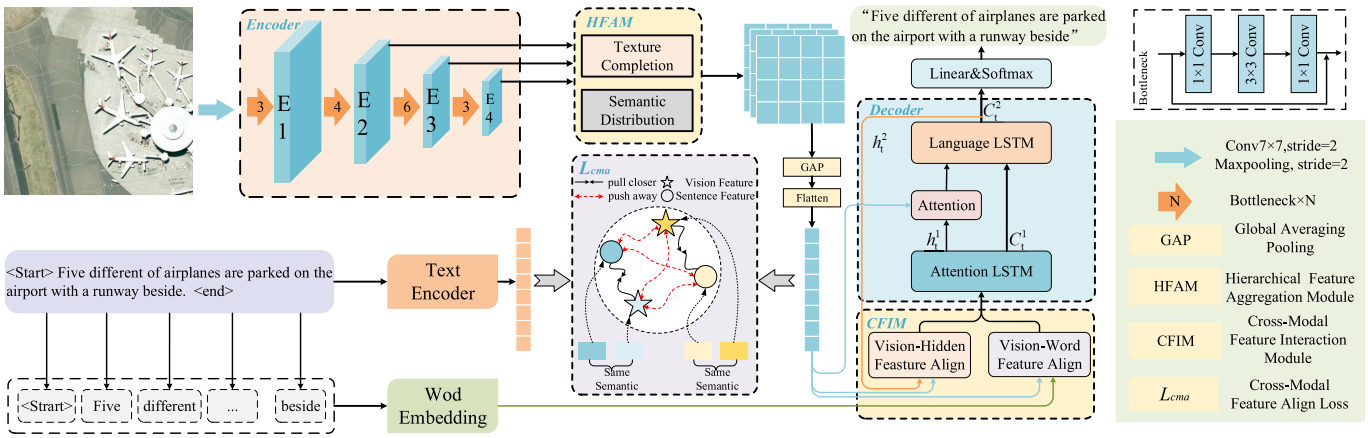


Fig. 1. Overall framework of the proposed HCNet, and it is divided into four parts: the encoder, the HFAM, the CFIM, and the dual-LSTM decoder.

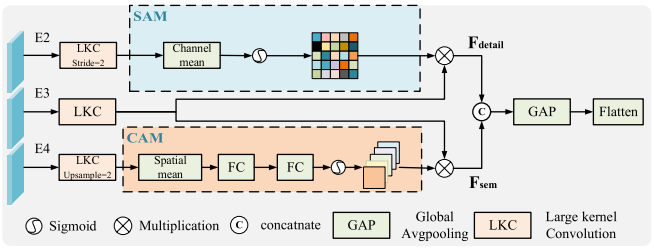


Fig. 2. Illustration of HFAM, where SAM means spatial attention mechanism and CAM means channel attention mechanism.

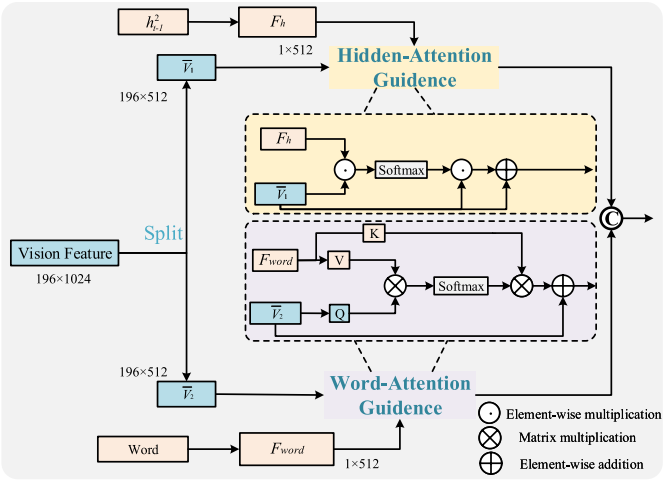


Fig. 3. Illustration of CFIM.

at the current time t. Since there is a remarkable feature disparity in remote sensing images in different modalities, only using concatenation or other simple operations to combine these inputs may not be sufficient to achieve an effective interaction. To address this trouble, a CFIM is designed, as shown in Fig. 3. Considering the different sources of input features, the module is mainly composed of two parts: *Hidden-Attention Guidance* and *Word-Attention Guidance*.

1) Hidden-Attention Guidance: The feature maps h_{t-1} are obtained by initializing the visual features $\overline{V}$, which contains text features generated before the current time. To align visual features with hidden states h_{t-1}, the hidden-attention guidance submodule is developed. First, we split the visual features into $\overline{V}_1 \in \mathbb{R}^{196\times512}$ and $\overline{V}_2 \in \mathbb{R}^{196\times512}$ based on the channel dimension. Then, the hidden features h_{t-1} are transformed into $F_h \in \mathbb{R}^{1\times512}$ by Squeeze and Linear operations. We conduct the multiplication operation remote sensing images $\overline{V}_1$ and F_h, and add a Softmax operation to produce channel weights. On this basis, the visual features $\overline{V}_1$ are weighted by multiplication operation along the channel dimension by the above weights. Finally, the input visual features and weighed visual features are combined through skip connections to generate the hidden state guided visual features F_{align1}. The overall process is defined as

$$F_{\text{align1}} = \text{Softmax}\left(\overline{V}_1 \odot F_h\right) \odot \overline{V}_1 + \overline{V}_1 \tag{6}$$

where $\odot$ is element-wise multiplication.

2) Word-Attention Guidance: To align between visual features and word features, the word attention guidance submodule is designed. Specifically, we employ Squeeze and Linear operations to transform the word features into $\overline{V}_2 \in \mathbb{R}^{1\times512}$. Then, the visual features $\overline{V}_2$ and generated word features F_{word} are fed into the multihead cross-attention mechanism, where $\overline{V}_2$ obtains the query features Q_i, and F_{word} yields the key features K_i and value features V_i, that is,

$$Q_i = \overline{V}_2 W_i^Q, \quad K_i = F_{\text{word}} W_i^K, \quad V_i = F_{\text{word}} W_i^V. \tag{7}$$

Here, i is the number of heads. W_i^Q, W_i^K, and W_i^V are learnable projection matrixes. Through a cross-attention mechanism, the visual features guided by word features are obtained. Finally, the visual features F_{align2} aligned with the word features is generated by combining the $\overline{V}_2$ and weighted visual features through skip connections, that is,

$$F_{\text{align2}} = \text{Concat}\left(\text{Softmax}\left(\frac{Q_i K_i^T}{\sqrt{d_k}}\right) V_i\right) + \overline{V}_2 \tag{8}$$

where d_k is the dimension of K_i. The visual features F_{align1} computed by hidden-attention guidance submodule and the visual vector F_{align2} produced by word-attention guidance submodule are combined and transformed to generate the visual features $F_{\text{align}} \in \mathbb{R}^{196\times1024}$, that is,

$$F_{\text{align}} = \text{Linear}\left(\left[F_{\text{align1}}, F_{\text{align2}}\right]\right). \tag{9}$$

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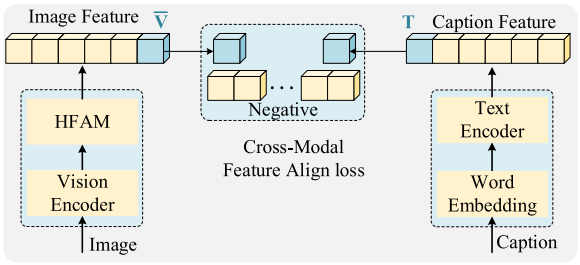


Fig. 4. Illustration of cross-modal feature align loss.

D. Dual-LSTM Decoder

Given the brief length of the remote sensing caption, we take dual-LSTM architecture [20] with Att-LSTM and L-LSTM as the decoder to decode in our approach. First, the aligned visual features F_{align} generated by the CFIM, the word feature F_{word} at the time t, and the hidden state h_{t-1}^2 output from the L-LSTM network are contacted and fed into the Att-LSTM to obtain the hidden states h_t^1 and c_t^1. The process is formulated as

$$h_t^1, c_t^1 = \text{Att-LSTM}\left(h_{t-1}^1, c_{t-1}^1, \left[h_{t-1}^2, F_{\text{word}}, F_{\text{align}}\right]\right). \quad (10)$$

Then, the attention mechanism is employed to refine the visual features, that is,

$$M = \text{Softmax}\left(W_a \tanh\left(W_v \overline{V} + W_h h_t^1\right)\right) \odot \overline{V}. \quad (11)$$

Subsequently, the refined visual features are merged with h_t^1 and input in the L-LSTM network to generate the hidden states h_t^2 and c_t^2, which is obtained by

$$h_t^2, c_t^2 = \text{L-LSTM}\left(h_{t-1}^2, c_{t-1}^2, \left[h_t^1, M\right]\right). \quad (12)$$

Finally, the next word is generated by performing the FC layer and Softmax operation on h_t^2, which projects h_t^2 into the word space, that is,

$$s_t^* = \text{Softmax}\left(\text{FC}\left(h_t^2\right)\right). \quad (13)$$

E. Loss Function

To train the model, we introduce two losses to optimize the RSIC model, that is, cross-entropy loss and cross-modal feature align loss. Specifically, the cross-entropy loss is defined as

$$\mathcal{L}_{\text{cap}} = -\sum_{t=1}^{T} \log\left(p_\theta\left(s_t^* | s_{1:t-1}^*\right)\right) \quad (14)$$

where $s_1^*, s_2^*, \ldots, s_{t-1}^*$ represent the label words. The difference between image and text features makes it challenging for the model to obtain high-quality captions. Inspired by contrastive language-image pretraining (CLIP) technology [40], the cross-modal feature align loss is introduced to align the global vision features $\overline{V}$ in HFAM with the semantic features T in the caption. As illustrated in Fig. 4, we utilize the LSTM as a text encoder to extract the sentence semantic information T from the caption. Then, the similarity between image–text pairs in a batch is computed by

$$\mathcal{L}_{I2T} = -\log \frac{\exp\left(\overline{V}_i \odot T_i\right)}{\sum_{j=1}^{n} \exp\left(\overline{V}_i \odot T_j\right)}$$
$$\mathcal{L}_{T2I} = -\log \frac{\exp\left(\overline{V}_i \odot T_i\right)}{\sum_{j=1}^{n} \exp\left(\overline{V}_j \odot T_i\right)} \quad (15)$$

where n represents the total number of captions in a batch. The cross-modal feature align loss is defined as

$$\mathcal{L}_{\text{cma}} = (L_{I2T} + L_{T2I})/2. \quad (16)$$

According to the above two losses, the total loss function of our method is

$$\mathcal{L}_{\text{loss}} = \lambda \mathcal{L}_{\text{cap}} + \mathcal{L}_{\text{cma}} \quad (17)$$

where λ is the balance factor. Finally, the weight is set to 3 in this article.

IV. Experiments

In this section, extensive experiments on the RSIC dataset are performed to demonstrate the effectiveness of HCNet. First, the dataset, evaluation metrics, and experiment settings are introduced. Then, a comparison and ablation study are conducted and analyzed. We also provide some visual results. Finally, we discuss the selection of hyperparameters in our experiments.

A. Datasets

The experiment uses Sydney, UCM, and NWPU caption datasets for evaluation. These datasets are derived from the scene classification datasets. The Sydney caption dataset contains 613 images with 0.5 m/pixel, and each image is annotated with five captions. The UCM caption dataset includes 2100 images with 0.3048 m/pixel and covers 21 scenes, where 10 500 sentences are available in this dataset. The NWPU caption dataset contains 31 500 images and 1 57 500 sentences. To the best of our knowledge, it is currently the largest dataset for RSIC. We follow [41], [42] to split the dataset, which aims to ensure a fair comparison.

B. Evaluation Metrics

1) BLEU [43]: The generated sentences are evaluated by calculating the match of n-grams between the predicted captions and ground truth (GT). Here, n-gram is a sequence of n continuous words with values ranging from 1 to 4.

2) METEOR [44]: It mainly evaluates machine translation task. Compared to the BLEU metric, METEOR considers word variations and synonyms of the same stem, which allows for more flexibility in calculating similarities.

3) ROUGE_L [45]: It measures the similarity by calculating the F-measure given the longest common subsequence between the generated captions and references.

4) CIDEr [46]: The metric is used to evaluate the image captioning task, which transforms the caption into a term frequency inverse-document frequency and weight n-gram. It focuses on the occurrence of keywords.

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5) Sm: Sm metric is calculated as the arithmetic mean of BLEU-4, METEOR, ROUGE_L, and CIDEr metrics. It can comprehensively evaluate the quality of generated captions.

C. Experimental Settings

As for the Sydney, UCM, and NWPU datasets, the images are resized to 224 × 224 pixels, and random flipping is employed to augment images during the training stage. The number of heads in CFIM is fixed to 8. The dimension of the word embedding is set to 512, and the hidden state dimensions of LSTM are set to 1000. During the training stage, the number of the count words to make vocabulary is set to 4, and the max length of the generated sentence is set to 25. We use Adam optimizer to optimize the model. The learning rate is set to 10^{-4} for Sydney and UCM datasets, and 5×10^{-4} for NWPU dataset. The batch size of the network is fixed to 64 and the maxepoch is set to 100. In the process of inference, we select the beam search value as 3, the random flipping and normalization are employed to augment images. All the experiments are implemented by PyTorch 1.10 in an NVIDIA RTX 3090.

D. Comparison With Existing Methods

To illustrate the effectiveness of the HCNet, several existing methods are selected and compared with the proposed model on three datasets. Table I shows that the proposed method achieves competitive performance on the Sydney dataset. Specifically, VLAD-LSTM utilizes hand-crafted features to generate a caption of the image, which leads to the model with limited representation ability. Therefore, it cannot obtain ideal values. GoogLeNet combined with soft/hard attention generates high-quality captions, which benefits deep exploration. It results in great performance. Considering the overfitting problem, TCE loss obtains better results on the Sydney dataset. In contrast, our HCNet yields competitive performance, especially in BLEU-4, ROUGE_L, and CIDEr metrics. In our view, there are two main reasons for this performance. The one is that the introduction of the HFAM attains accurate and complete visual features. It facilitates the model in generating accurate descriptions. The other is that the proposed CFIM and loss $\mathcal{L}_{\text{cma}}$ allow the model to better align different modal features. This can produce coherent captions in terms of language.

As seen in Table II, our method achieves the best performance for almost all metrics on the UCM dataset. Concretely, the BLEU-4 obtained by the proposed method reaches 74.49%, which is 7.49% higher than that of TCE loss. Meanwhile, our method yields 0.9% improvement on METEOR compared with TCE loss, which shows the generated captions that cover important information in the reference labels. Unlike the results on the above dataset, GVFGA + LSGA shows the highest ROUGE_L and CIDEr values, which reveals that it has poor robustness. The Purt model achieves a competitive performance in the UCM dataset, which is because the model is entirely based on the Transformer for extracting image features and generating descriptions, thus minimizing the cross-modal feature differences is vital for the RSIC task. Meanwhile, the RS-CapRet model based on the LLM technique achieves the best Cider, demonstrating the potential of the LLM for the RSIC task.

The above experimental results prove that our method exhibits significant advantages in comprehensive performance over existing methods, particularly in dealing with large-scale datasets. Similarly, our HCNet also demonstrates superior performance on the NWPU dataset in Table III, especially in the case of the Cider metric, which outperformed the Purt model by 14.15%. The comparison experiment indicates the proposed method can address these scenes across datasets well.

E. Ablation Study

In this section, we conduct an ablation experiment on the UCM dataset to demonstrate the influence of HFAM, CFIM, and cross-modal feature align loss function. Here, the combination of different parts is defined as a Model.

1) Effect of HFAM: The proposed HFAM aggregates features at different scales in Model2 achieves significant improvements across all metrics compared with the Baseline (ResNet50 + dual-LSTM) in Table IV. These results provide evidence that the aggregation of multiresolution features enables the model to acquire comprehensive feature representations. As a result, the model performs better in object classification and greatly improves the accuracy of generated captions. Furthermore, we also consider the feature fusion strategy [13], which adopts multiple operations to fuse features with different resolutions. Compared with Model 3, the Baseline combined with HFAM has a better performance. This improvement can be attributed to the detail and semantic information.

2) Effect of CFIM: Compared with Baseline, Model 4 with CFIM increases 1.27% in ROUGE_L. This improvement enhances the alignment effect of cross-modal features, which facilitates an effective interaction of visual and textual features in the decoder. CFIM enables the LSTM to generate accurate hidden states, which is conducive to guiding accurate caption generation. Therefore, the generated sentences are closer to the ground truth. In contrast with Model 2 and Model 3, Model 5 demonstrates improvements in most metrics when HFAM and CFIM are embedded in the model. These experiment results indicate that the integration of designed modules yields higher-quality captions.

3) Effect of the $\mathcal{L}_{cma}$ Loss Function: To minimize the difference between image and text features, cross-modal align loss function is introduced in Model 5. Cross-modal features with the same semantics are further aligned under the guidance of this loss function to improve the performance of predicted captions. Therefore, our method obtains the highest values in all metrics, which reflects that the captions generated by HCNet are closer to the references. This experiment demonstrates the importance of incorporating the cross-modal align loss function for the RSIC task.

F. Visual Results

To quantitatively show the generation process in captions, specific words, sentences, and corresponding attention maps

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TABLE I

EXPERIMENTS RESULTS ON THE SYDNEY DATASET. THE BOLD AND UNDERLINED INDICATE THE BEST AND SECOND BEST RESULTS. THE ∗ DENOTES THE RESULTS OF OUR REIMPLEMENTED

Method	BLEU-I↑	BLEU-2↑	BLEU-3↑	BLEU-4↑	METEOR↑	ROUGE_L↑	CIDEr↑	Sm↑
VLAD-LSTM [31]	0.4913	0.3412	0.2760	0.2314	0.1930	0.4201	0.9164	0.4402
Sound-a-a [36]	0.7093	0.6228	0.5393	0.4602	0.3121	0.5974	1.7477	0.7794
Multimodal [41]	0.6980	0.6130	0.5440	0.5050	0.3610	0.6370	2.2020	0.9262
Soft Attenttion [31]	0.7128	0.6239	0.5527	0.4924	0.3675	0.6913	2.0343	1.0471
Hard Attention [31]	0.7689	0.6613	0.5840	0.5170	0.3719	0.6842	1.9863	0.9541
TCE loss [35]	0.7937	0.7304	**0.6717**	**0.6193**	**0.4430**	0.7130	2.4042	1.0449
Word-Sentence framework [32]	0.7891	0.7094	0.6317	0.5625	0.4181	0.6922	2.0411	0.9285
GVFGA+LSGA [15]	0.7681	0.6846	0.6145	0.5504	0.3866	0.7030	2.4522	1.0231
SVM-D BOW [33]	0.7787	0.6835	0.6023	0.5305	0.3797	0.6992	2.2722	0.9704
SVM-D CONC [33]	0.7547	0.6711	0.5970	0.5308	0.3643	0.6746	2.2222	0.9480
MLCANet [42]	**0.8310**	**0.7420**	0.6590	0.5800	0.3900	0.7110	2.3240	1.0012
MLAT∗ [13]	0.7768	0.7034	0.6422	0.5862	0.3882	0.7167	2.3605	1.0129
Post-processing [47]	0.7837	0.6985	0.6322	0.5717	0.3949	0.7106	**2.5553**	**1.0581**
RS-CapRet [50]	0.7870	0.7000	0.6280	0.5640	0.3880	0.7070	2.3920	1.0127
HCNet	0.7686	0.7109	0.6573	0.6102	0.3980	**0.7172**	2.4714	1.0492

TABLE II

EXPERIMENTS RESULTS ON THE UCM DATASET. THE BOLD AND UNDERLINED INDICATE THE BEST AND SECOND BEST RESULTS. ∗ DENOTES THE RESULTS OF OUR REIMPLEMENTED

Method	BLEU-I↑	BLEU-2↑	BLEU-3↑	BLEU-4↑	METEOR↑	ROUGE_L↑	CIDEr↑	Sm↑
VLAD-LSTM [31]	0.7016	0.6085	0.5496	0.5030	0.3464	0.6520	2.3131	0.9536
Sound-a-a [36]	0.7484	0.6837	0.6310	0.5896	0.3623	0.6579	2.7281	1.0845
Multimodal [41]	0.7100	0.5980	0.5530	0.4600	0.3430	0.6620	2.9260	1.0977
Soft Attenttion [31]	0.7454	0.6545	0.5855	0.5250	0.3886	0.7237	2.6124	1.0624
Hard Attention [31]	0.8157	0.7312	0.6702	0.6182	0.4263	0.7698	2.9947	1.2023
TCE loss [35]	0.8210	0.7622	0.7140	0.6700	0.4775	0.7567	2.8547	1.1897
Word-Sentence framework [32]	0.7931	0.7237	0.6671	0.6202	0.4395	0.7132	2.7871	1.1400
GVFGA+LSGA [15]	0.8319	0.7657	0.7103	0.6596	0.4436	0.7845	3.3270	1.3037
SVM-D BOW [33]	0.7635	0.6664	0.5869	0.5195	0.3654	0.6801	2.7142	1.0717
SVM-D CONC [33]	0.7653	0.6947	0.6417	0.5942	0.3702	0.6877	2.9228	1.1437
MLCANet [42]	0.8260	0.7700	0.7170	0.6680	0.4350	0.7720	3.2400	1.2787
MLAT∗ [13]	0.8226	0.7539	0.6959	0.6424	0.4268	0.7729	3.0826	1.2311
Post-processing [47]	0.7973	0.7298	0.6744	0.6262	0.4080	0.7406	3.0964	1.2186
Clipcap ∗ [50]	0.8213	0.7488	0.6872	0.6320	0.4232	0.7742	3.2391	1.2671
PureT [26]	0.8573	0.8020	0.7562	0.7129	0.4686	0.8201	3.4900	1.3729
RS-CapRet [51]	0.8430	0.7790	0.7220	0.6700	0.4720	0.8170	**3.5480**	1.3767
HCNet	**0.8826**	**0.8335**	**0.7885**	**0.7449**	**0.4865**	**0.8391**	3.5183	**1.3972**

TABLE III

EXPERIMENTS RESULTS ON THE NWPU DATASET. THE BOLD AND UNDERLINED INDICATE THE BEST AND SECOND BEST RESULTS. ∗ DENOTES THE RESULTS OF OUR REIMPLEMENTED

Method	BLEU-I↑	BLEU-2↑	BLEU-3↑	BLEU-4↑	METEOR↑	ROUGE_L↑	Cider↑	Sm↑
Multimodal [41]	0.7250	0.6030	0.5180	0.4550	0.3360	0.5910	1.1790	0.6402
Soft Attention [31]	0.7310	0.6090	0.5250	0.4620	0.3390	0.5990	1.1360	0.6340
Hard Attention [31]	0.7330	0.6100	0.5270	0.4640	0.3400	0.6000	1.1030	0.6267
FC-Att+LSTM [48]	0.7360	0.6150	0.5320	0.4690	0.3380	0.6000	1.2310	0.6595
SM-Att+LSTM [48]	0.7390	0.6170	0.5320	0.4680	0.3300	0.5930	1.2360	0.6567
MLCANet [42]	0.7540	0.6240	0.5410	0.4780	0.3370	0.6010	1.2640	0.6700
BUTD∗ [20]	0.8718	0.7878	0.7206	0.6669	0.4304	0.7612	1.8866	0.9362
AoANet∗ [49]	0.8755	0.7754	0.6917	0.6207	0.3907	0.7301	1.849	0.8976
MLAT∗ [13]	0.8527	0.7674	0.7007	0.6479	0.4327	0.7496	1.8556	0.9214
Clipcap∗ [50]	0.8394	0.7421	0.6619	0.5954	0.4142	0.7385	1.7265	0.8685
PureT [26]	0.8880	0.8031	0.7330	0.6750	0.4232	0.7584	1.9512	0.9519
RS-CapRet [51]	0.8710	0.7870	0.7170	0.6560	0.4360	0.7760	1.9290	0.9492
HCNet	**0.8954**	**0.8251**	**0.7658**	**0.7168**	**0.4742**	**0.8053**	**2.0927**	**1.0222**

are selected. Fig. 5 depicts the examples of sentences generated by HCNet. This figure lists three distinct scenes, including an airplane, a baseball diamond, and a building. In the first sample, our approach generates the word "beige," which closely resembles the color "white." Throughout the description process, the model notably pays more attention to

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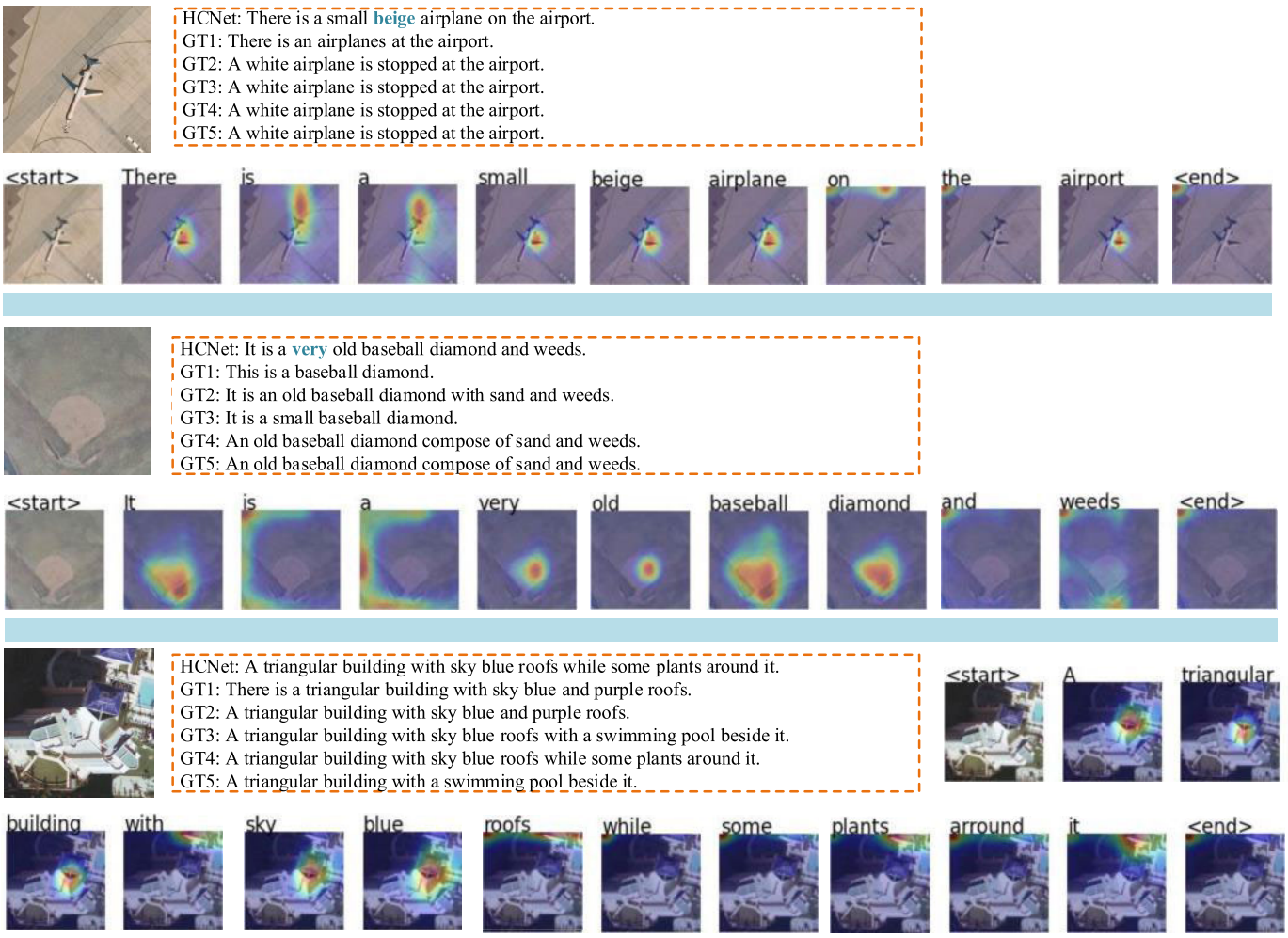


Fig. 5. Visual results of each word in the generated sentence. The closer the color is to red means that the model pays more attention to this area. The blue indicates the difference between our results and GT.

TABLE IV

ABALTION RESULTS ON THE UCM DATASET. THE BOLD AND UNDERLINED INDICATE THE BEST AND SECOND BEST RESULTS

Components						Metric							
Method	Baseline	HFAM	R&Cat [13]	CFIM	$\mathcal{L}_{cma}$	BLEU-1 ↑	BLEU-2 ↑	BLEU-3 ↑	BLEU-4↑	METEOR↑	ROUGE_L↑	Cider ↑	Sm ↑
Model1	✓					0.8392	0.7778	0.7273	0.6820	0.4486	0.7895	3.1416	1.2654
Model2	✓	✓				0.8658	0.8148	0.7690	0.7273	<u>0.4855</u>	0.8352	3.4343	1.3706
Model3	✓		✓			0.8597	0.8067	0.7584	0.7136	0.4608	0.8091	3.3437	1.3318
Model4	✓			✓		0.8429	0.7793	0.7263	0.6796	0.4433	0.8022	3.1529	1.2695
Model5	✓	✓		✓		<u>0.8805</u>	<u>0.8270</u>	<u>0.7796</u>	<u>0.7361</u>	0.4779	<u>0.8355</u>	<u>3.4953</u>	<u>1.3862</u>
HCNet	✓	✓		✓	✓	**0.8826**	**0.8335**	**0.7885**	**0.7449**	**0.4865**	**0.8391**	**3.5183**	**1.3972**

the location area of the aircraft, when generating words related to aircraft, such as “a,” “small,” “airplane,” and “airport.” In the second sample, the model predicts the word “very,” which enhances the coherence of the sentence. In the third sample, the model focuses on the corresponding location area in the image while generating the words “building” and “roofs.” These samples presented in this figure demonstrate that our HCNet can provide precise descriptions of objects in the RSI.

To analyze the experimental results, several predicted results are selected to show. Fig. 6 displays five examples on the Sydney, UCM, and NWPU datasets. These contents include experimental images, corresponding manual annotations (five in total), and image descriptions generated by Baseline and OurNet. Specifically, As the images (a) and (b) show on the Sydney dataset, the Baseline can identify “Many houses” in the image but fails to provide a comprehensive depiction of the global features. In contrast, our method effectively integrates multiscale image features, which enhances the ability of feature representation. Consequently, it generates the “A residential area.” As shown in image (c), it is obvious that the description of “An empty of with” predicted by the Baseline contains a grammatical error. Different from this, the CFIM

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Fig. 6. Examples of sentences generated by HCNet from the Sydney dataset [first row, (a)–(e)], the UCM dataset [second row, (f)–(j)], and the NWPU dataset [third row, (k)–(o)]. The red words and blue words indicate the mistakes and advantages of our method compared to the baseline, respectively.

designed by our approach can effectively integrate visual and text features, which reduces the appearance of grammatical errors. As for image (d), our model demonstrates superior recognition ability, whereas the Baseline wrongly identifies "river" in the image as "lawn." Similarly, the proposed model obtains more comprehensive captions compared with the

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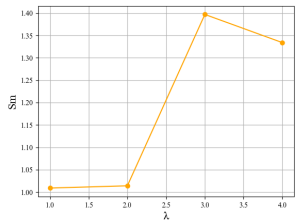


Fig. 7. Comparison results with the different value of λ in the UCM dataset.

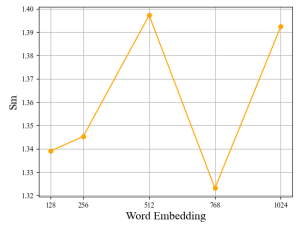


Fig. 8. Comparison results with the different values of word embedding in the UCM dataset.

Baseline in image (e). Furthermore, we also show the results on UCM and NWPU datasets. The experiments exhibit that our scheme can achieve higher accuracy and completeness of the captions.

G. Hyperparameter Selection

In this section, we conduct a hyperparameter selection experiment on the UCM dataset to demonstrate the influence of λ and the dimension of Word Embedding.

1) Value of λ: The weight of the loss function is an important hyperparameter in the model optimization phase. It is difficult to align the visual features obtained by the encoder with the same semantic information of the text features extracted by captions while the value is small. Conversely, a large value introduces optimization bias within the framework, impeding the generation of high-quality text descriptions. As shown in Fig. 7, we set λ from 1 to 4, and when the value is set to 3, the model achieves the best value in Sm metric, which proves that the model delivers the best comprehensive performance under this configuration.

2) Dimension of Word Embedding: The dimension of word embedding stands as a crucial parameter in caption modeling. The model with a small value of dimension fails to capture enough semantic information embedded in the text, and when the value is large, it decreases the representation ability of visual features. As shown in Fig. 8, when the value is set to 512, the model obtains the highest score.

V. Conclusion

This article proposes an RSIC approach that leverages hierarchical feature aggregation and cross-modal feature alignment. To effectively utilize the features of different scales, the hierarchical feature aggregation module is designed to enhance feature representation. Additionally, the CFMI is developed to align visual features with different input features in the decoder, which facilitates the model to generate accurate hidden states. Considering the disparities between the two modal features, we introduce the cross-modal feature align loss to align vision features and text features, thus increasing the overall quality of generated captions. The experiments are conducted on three public datasets, and the results demonstrate the superiority of the proposed method in terms of quantitative and qualitative aspects. At present, the large model shows great advantages in computer vision. In the future, we will combine large model technology to improve the performance of the RSIC model.

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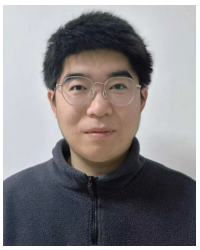
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